A Group-theoretic view of Music

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"Music is the one incorporeal entrance into the higher world of knowledge which comprehends mankind but which mankind cannot comprehend"

- Ludwig van Beethoven

"The universe is an enormous direct product of representations of symmetry groups"

- Hermann Weyl

Abstract:
The purpose of this paper is to study basic music theory using notions of abstract algebra, more precisely group theory. In the first two sections, the needed background of both music and group theory is presented and in the last section, the coexistence of the two is analyzed, mainly by the works of Hook ([2]), Lewin ([10]) and Riemann.

1 Introduction

1.1 Some words on Music and Mathematics

Anybody studying or even listening to music, can recognize some patterns or motivos, which seem to have a sense of symmetry. Truth is, there exists a deep mathematical structure underlying our favorite pieces. The system of music theory itself is based on connections between various intervals and chords with each other, which can be compared to many mathematical systems. The mathematical thinking is somehow necessary for someone who wants to understand basic music theory. This can be verified even from the ancient Greeks[1]. The Pythagoreans noticed that three particular intervals where the most aesthetically pleasing, which now we call an octave, a fifth and a fourth. So they were the first to relate musical intervals with number theory, more specifically rational numbers, by measuring the length of a vibrating string. The fraction denoted the frequency of the higher note to that of the lower.

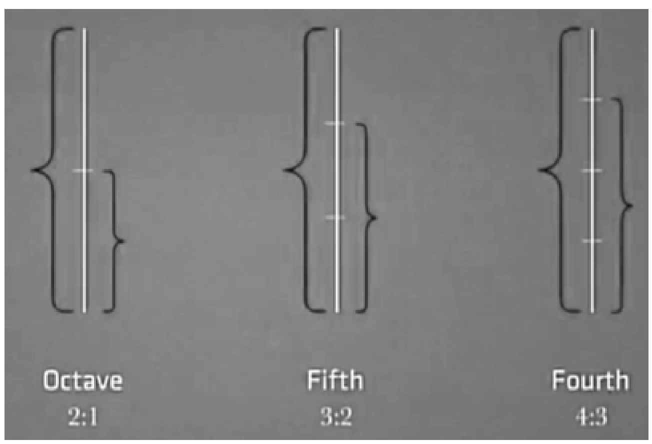


Figure 1: For an octave the ratio is 2:1, for a fifth 3:2 and for a fourth 4:3

The question of which ratios sound harmonious caught the attention of many known scientists, mainly around the 18th century (the so-called ”Age of Enlightenment”).

Soon after that, about the starts of the 19th century was the first appearance of groups in mathematics. The fact that I referenced the word ”symmetry” above when referring to music, is no coincidence. Group theory is at its core the study of symmetry between, not only abstract mathematical ideas, but also real objects. A group may be one of the central themes in abstract algebra, but it describes reality in a very precise way, examples of that varying from a simple puzzle such as the Rubik's cube to the crystals in a molecule or

[1]such as Plato, Pythagoras, Philolaus, Archytas.

even the standard model of particles in the universe. Although the interest of this paper is the connection between music theory and group theory, there exist many more modern mathematical interpretations of music theory, as in [4] and [6] where a general notion of continuity is applied, giving it the chance to be studied geometrically, in contrast with the context here, where the approach is more discrete.

The main idea of constructing a purely algebraic structure which would preserve both the mathematical and musical axioms came from J.Hook in [2] (2002), where he introduced the Uniform Triadic Transformations (UTTs) which generalized some preexisting ideas. Another great example is H.Riemann's Neo-Riemannian theory, which also focuses on transformations and their applications in chord progressions and harmony.

2 Music theory

One can identify music theory as the study of intervals between pitches and their combinations so the result is "aesthetically pleasing" or "harmonious"[2]. Important notions that come with musical intervals, which we will define in this section, are chords, rhythm, melody, scales and modes, dynamics, articulation and so on. Usually, after music theory is learned, one proceeds to Harmony which involves studying the rules applied in chord progressions. One can think of Harmony as a horizontal study of music, in contrast with music theory.

We may start by defining what a note and a pitch is.

Definition 2.1. A **note** is the smallest bit of sooound there is. Usually notes are named as "Do, Re, Mi, Fa, Sol, La, Si" or "C, D, E, F, G, A, B" respectively.[3]

Definition 2.2. Pitch is the object which describes the frequency a musical note is producing, i.e. how low or high that frequency is. It is measured in Hertz. Usually, notes with the same pitch are assigned the same letter, going from lowest to the highest frequency: "A, B, C, D, E, F, G".

Now, we can give a second more precise definition of a note.

Definition 2.3. A **note** is the pitch and duration of a sound. It is usually denoted using musical notation[4].

Definition 2.4. A **chord** is a set of three or more notes played simultaneously.

This is denoted as following:

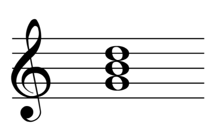


Figure 2: A G major chord

[2]This is the reason that many believe music theory to be subjective
[3]seen how they are arranged on a piano in Appendix B
[4]Appendix A

Definition 2.5. Rythm (or **Tempo**) is the arrangement of sounds and silences in a musical piece.

We usually denote rythm in the beginning of the pentagram as a ratio[5], which divides the piece into meters. In the following example, the rhythm is 3/4 which roughly means that the sum of all the musical beats of every meter is 3.

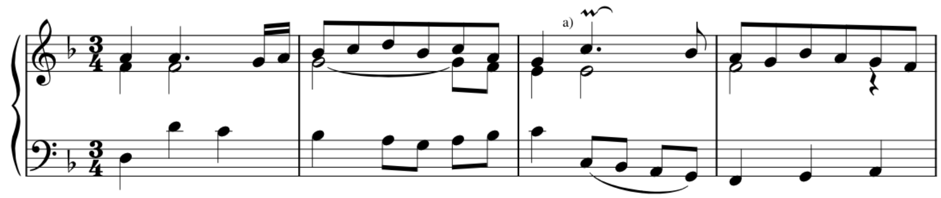


Figure 3: From Bach's French Suite no.1

Now we can proceed to the most important part of music theory, musical intervals.

Definition 2.6. A musical **interval** is the distance between two pitches.

The first musical interval one can measure is the **semitone**, which is the closest one note can ascend or descend. In the piano, this is the two notes "touching" it from the right and from the left.

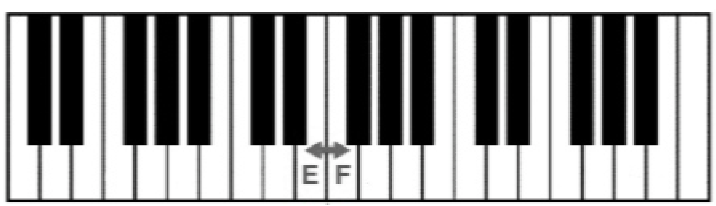


Figure 4: For example, E and F are a semitone appart

Before revealing what musical intervals there are, we must first define three unary operations that can act on a note.

Definition 2.7. The **sharp** denoted by "$\sharp$", when acted on a note it shifts it one semitone to the right (or up).

Definition 2.8. The **flat** denoted by "$\flat$", when acted on a note it shifts it one semitone to the left (or down).

In that notion, $E^\sharp$ is the same as F and $F^\flat$ is the same as E. Notes such as these, that have different notation but sound the same are called **enharmonically equivalent**.

Definition 2.9. The **natural** denoted by "$\natural$", when acted on a note which already has been shifted by a semitone, it returns it to its original place.

[5]Sometimes, we write a c instead of 4/4

When these symbols, which we usually refer as accidentals, appear at the beginning of the pentagram, before the rhythm, it means that they operate on any note that they are placed on, i.e. in Figure 3 every B which is played is shifted a semitone down.

We can now write all the musical intervals in the following table, assigning to them the number of semitones they contain, which is wildly important for later:

Table 1

Name	Number of Semitones	Notation
Perfect First	0	P1
Minor Second/Semitone/Half Step	1	m2
Major Second/Whole Tone/Whole Step	2	M2
Minor Third	3	m3
Major Third	4	M3
Perfect Fourth	5	P4
Diminished Fifth/Augmented Fourth	6	d5/A4
Perfect Fifth	7	P5
Minor Sixth	8	m6
Major Sixth	9	M6
Minor Seventh	10	m7
Major Seventh	11	M7
Perfect Octave	12	P8

To lay out some examples, $C^\sharp E = m3$, $GD^\flat = d5$, $AF = m6$, $B^\flat C = M2$ and so on.

Definition 2.10. Pitches that differ by an integer multiple of an octave, or are enharmonically equivalent belong in the same **pitch-class**.

When having the various notes with various pitches, one can arrange them with respect to their pitch and form what we call a scale.

Definition 2.11. A musical **scale** is a set of notes arranged by an ordered pitch.

For example, a scale in which its notes have increasing pitch is an ascending scale and vice versa.

There are two kinds of scales, that differ by the number of semitones (or half steps) and tones (or whole steps) that they have.

Definition 2.12. A **major** scale has its whole steps and half steps arranged as:

$$W\,W\,H\,W\,W\,W\,H$$

Definition 2.13. A **minor** scale has its whole steps and half steps arranged as:

$$W\,H\,W\,W\,H\,W\,W$$

Generally, the major scale is known to sound "happy" in contrast to the minor scale which sounds "sad". The scales are defined by the accidentals used in the beginning of the pentagram. For example, again in Figure 3, the suite is in a D minor scale which contains the pitches: $D\,E\,F\,G\,A\,B^{\flat}\,C, D$ for the whole step and half step rule to be satisfied, beginning from D. This is the reason we have a $B^{\flat}$ after the rhythm, to indicate that all the B's that are played will be "flated".
We can also distinguish a major from a minor chord from the interval of the first two notes. As major chords have a major third in contrast with the minor chords that have a minor third.

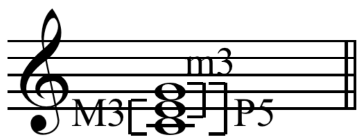


Figure 5: A C major chord

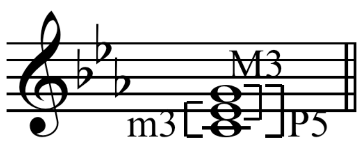


Figure 6: A C minor chord

Therefore, by changing the third from major to minor or vice versa, we can change the whole chord, making a radical difference to the sound outcome.

3 Group theory

3.1 The basic notions of groups

A recurring theme in modern mathematics is taking a set (i.e. a collection of elements) and equipping it with some additional structure. Dependent on the structure we equip, some very interesting properties emerge. A wildly known example of such an object is a group. Having a set G one can equip it with a map, or an operation [6] which is closed under G and then demand four axioms to be satisfied. That defines a group.

Definition 3.1. A **group** G is a set of elements together with a binary map, called the group operation $\bullet : G \times G \longrightarrow G$, that satisfy the following axioms:

1. Closure: $\forall\ a, b \in G : a \bullet b \in G$
2. Associativity: $\forall\ a, b, c \in G : (a \bullet b) \bullet c = a \bullet (b \bullet c)$
3. Identity[7] element: $\exists!\ e \in G : e \bullet a = a \bullet e = a\ \ \forall\ a \in G$
4. Inverse element: $\forall\ a \in G\ \ \exists!\ a^{-1} \in G : a \bullet a^{-1} = a^{-1} \bullet a = e$

Definition 3.2. A group $(G, \bullet)$ is called **abelian**[8], if all its elements commute under the group operation:

$$\forall\ a, b \in G : a \bullet b = b \bullet a$$

Definition 3.3. A **subgroup** of a group $(G, \bullet)$ is a subset of G, which itself must satisfy the four axioms, thus forming a group under the group operation of G.

Examples:

(a) Both the integers and rational numbers form abelian groups with identity element $e = 0$ under each addition respectively: $(\mathbb{Z}, +_{\mathbb{Z}})$, $(\mathbb{Q}, +_{\mathbb{Q}})$.

(b) The real numbers without zero $\mathbb{R} \setminus \{0\} =: \mathbb{R}_{\neq 0}$, form an abelian group with identity element $e = 1$ under multiplication: $(\mathbb{R}_{\neq 0}, \cdot_{\mathbb{R}})$.

(c) The complex numbers without zero $\mathbb{C}\setminus\{0\} =: \mathbb{C}_{\neq 0}$ form an abelian group with identity element $e = 1 + 0i = 1$ under complex multiplication: $(\mathbb{C}_{\neq 0}, \cdot_{\mathbb{C}})$.

Also one can see them as subgroups: $(\mathbb{Z}, +_{\mathbb{Z}}) \subset (\mathbb{Q}, +_{\mathbb{Q}}) \subset (\mathbb{R}_{\neq 0}, \cdot_{\mathbb{R}}) \subset (\mathbb{C}_{\neq 0}, \cdot_{\mathbb{C}})$.

(d) The set $\mathbb{S}^1 = \{z \in \mathbb{C} \,|\, |z| = 1\}$ forms an abelian group with identity element $e = 1$ under complex multiplication, as a subgroup of the complex numbers: $(\mathbb{S}^1, \cdot_{\mathbb{C}})$.[9]

[6] generaly called a "product"
[7] or neutral
[8] or commutative
[9] This group plays an important role in theoretical physics and its usually called the unitary group $U(1)$.

(e) The set $GL(n,\mathbb{R}) := \{\mathbb{A} \mid \mathbb{A} \text{ is an } n\times n \text{ matrix: } det(\mathbb{A}) \neq 0\}$ of all the $n \times n$ real matrices with non-vanishing determinant forms a non-abelian group under matrix multiplication, since its not commutative, with identity element the unitary matrix $e = \mathbb{I}_{n\times n}$: $(GL(n,\mathbb{R}),\cdot)$.

Definition 3.4. A group G is called **cyclic** if it can be generated by some element $g \in G$, that is applying the group operation on g can give the whole group. Denoted by $< g >:= \{g \bullet g \bullet \ldots \bullet g \equiv g^n \mid n \in \mathbb{N}\} = G$.

Theorem 3.1. *Every cyclic group is abelian.*

Proof. If G is a cyclic group, then there is a $g \in G$ which generates it. So if we take two arbitraty elements $a, b \in G$, we can write them as: $a = g^n$ and $b = g^m$, for some $n, m \in \mathbb{N}$. Then,

$$a \bullet b = g^n \bullet g^m = g^m \bullet g^n = b \bullet a$$

since g commutes with itself. ■

Definition 3.5. The **order** of a group G is its cardinality $|G|$.

Theorem 3.2. *Let $(G,\bullet_G)$ and $(K,\bullet_K)$ be two groups. The set $G\times K := \{(g,k) \mid g \in G$ and $k \in K\}$ also forms a group $(G \times K,\cdot)$, the **direct product** [10] group of G and K with the group operation being component-wise multiplication:*

$$(g_1,k_1)\cdot(g_2,k_2) := (g_1 \bullet_G g_2, k_1 \bullet_K k_2)$$

Proof.
1. Closure: $\forall (g_1,k_1),(g_2,k_2) \in G\times K : (g_1 \bullet_G g_2, k_1 \bullet_K k_2) \in G \times K$.

2. Associativity: Both G and K are associative as groups, therefore the associativity holds.

3. The identity element of the $(G\times K,\cdot)$ is (e_G, e_K).

4. The inverse element of any $(g,k) \in G\times K$ is (g^{-1}, k^{-1}).

Finally, if G and K are abelian, then $G \times K$ is also abelian. ■

<u>Examples</u>:

(a) The $n-$dimensional Eucledian space, equiped with point-wise addition is an abelian group $(\mathbb{R}^n,+)$. This actually is the direct product of $\mathbb{R}$ with itself $n-$times: $(\underbrace{\mathbb{R}\times\mathbb{R}\times\cdots\times\mathbb{R}}_{\text{n-times}},+)$.

(b) The group $(\mathbb{C}_{\neq 0}, +_{\mathbb{C}})$ is actually the direct product of the unit complex circle and the positive real numbers $\mathbb{R}_{>0} := \{x \in \mathbb{R} \mid x > 0\}$ under multiplication: $(\mathbb{S}^1 \times \mathbb{R}_{>0},\cdot)$.[11] □

Also, for the order of the direct product group, we have: $|G\times K| = |G||K|$.

[10] Sometimes denoted as $G \bigoplus K$

[11] If interested in the proof, see Appendix C

3.2 Structure-preserving maps for groups

When having a mathematical object, one is usually interested in maps that preserve its structure which are called **morphisms**. In this section, we will present the basic morphisms needed for group theory.

Definition 3.6. Let $(G, \bullet_G)$ and $(K, \bullet_K)$ be groups. A **group Homomorphism** is a map $f : G \to K$ that preserves the group product. That is $\forall g_1, g_2 \in G : f(g_1 \bullet_G g_2) = f(g_1) \bullet_K f(g_2)$.

Note that the product inside the map is the product on G and the product on the right side is the product on K.

Examples:

(a) Consider the two groups: $(\mathbb{Z}, +_{\mathbb{Z}})$ and $(\{-1, 1\}, \cdot_{\mathbb{Z}})$. We can construct the map $f : \mathbb{Z} \to \{-1, 1\}$ as:

$$x \mapsto \begin{cases} 1 & \text{if } x \text{ is even} \\ -1 & \text{if } x \text{ is odd} \end{cases}$$

If we choose $x, y \in \mathbb{Z}$ both even, they can be written as $x = 2n$ and $y = 2m$ for m, n: odd. Then:

$$f(x +_{\mathbb{Z}} y) = f(2n +_{\mathbb{Z}} 2m) = f(2(n +_{\mathbb{Z}} m))$$

And let $n +_{\mathbb{Z}} m := l_1$ which is even, so is $2l_1$:

$$\Rightarrow f(2l_1) = 1 = f(x) \cdot_{\mathbb{Z}} f(y)$$

If we choose $x, y \in \mathbb{Z}$ where x is even and y is odd, they can be written as $x = 2n$ and $y = m$ for m, n: odd. Then:

$$f(x +_{\mathbb{Z}} y) = f(2n +_{\mathbb{Z}} m)$$

And let $2n +_{\mathbb{Z}} m := l_2$ which is odd, so:

$$\Rightarrow f(l_2) = -1 = 1 \cdot_{\mathbb{Z}} (-1) = f(2n) \cdot_{\mathbb{Z}} f(m) = f(x) \cdot_{\mathbb{Z}} f(y)$$

If we choose $x, y \in \mathbb{Z}$ both odd, then:

$$f(x +_{\mathbb{Z}} y) = 1 = (-1) \cdot_{\mathbb{Z}} (-1) = f(x) \cdot_{\mathbb{Z}} f(y)$$

So f preserves the group operation, thus is a homomorphism. $\square$

(b) Consider the groups $(GL(2, \mathbb{R}), \cdot)$ and $(\mathbb{R} \setminus \{0\}, \cdot_{\mathbb{R}})$. Now we define the map $f : GL(2, \mathbb{R}) \to \mathbb{R} \setminus \{0\}$ as:

$$f(\mathbb{A}) = det(\mathbb{A})$$

$\forall\ \mathbb{A}, \mathbb{B} \in GL(2, \mathbb{R})$:

$$f(\mathbb{A} \cdot \mathbb{B}) = det(\mathbb{A} \cdot \mathbb{B}) = det(\mathbb{A}) \cdot_{\mathbb{R}} det(\mathbb{B}) = f(\mathbb{A}) \cdot_{\mathbb{R}} f(\mathbb{B})\ \square$$

(c) Consider the two groups: $(\mathbb{C}, +_{\mathbb{C}})$ and $(\mathbb{C} \setminus \{0\}, \cdot_{\mathbb{C}})$. We can define the complex exponential map $f : \mathbb{C} \to \mathbb{C} \setminus \{0\}$ as:

$$z \mapsto exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

We can almost trivialy show that this is a group homomorphism from the property of the exponential sum

$$exp(z_1 +_{\mathbb{C}} z_2) = exp(z_1) \cdot_{\mathbb{C}} exp(z_2) \ \square$$

Definition 3.7. An **Isomorphism** is a bijective[12] Homomorphism.

This is the notion of equality in set theory, as we take two isomorphic sets A and B, meaning there exist an isomorphism from A to B, to be the same. One can think of it as pairing up the elements of A and B. Denoted by $A \cong B$.

Theorem 3.3. *Any isomorphism maps the identity element of one group into the identity element of the other.*

Proof. We have the isomorphism $f : G \to K$. If we choose the two elements of G to be $g_1 = e_G$ and $g_2 = e_G$, the property gives:

$$f(e_G \bullet_G e_G) = f(e_G) \bullet_K f(e_G)$$

$$\Rightarrow f(e_G) = f(e_G) \bullet_K f(e_G)$$

Now multiplying with the product of K by $f^{-1}(e_G)$ from the left to both sides:

$$\Rightarrow f^{-1}(e_G) \bullet_K f(e_G) = f^{-1}(e_G) \bullet_K f(e_G) \bullet_K f(e_G)$$

$$\Rightarrow e_K = e_K \bullet_K f(e_G)$$

$$\Rightarrow e_K = f(e_G) \ \blacksquare$$

Examples:

(a) The groups $(\mathbb{Z}, +_{\mathbb{Z}})$ and $(2\mathbb{Z}, +_{\mathbb{Z}})$ where $2\mathbb{Z} := \{n \in \mathbb{Z} \,|\, n = 2m \text{ for } m \in \mathbb{Z}\}$ is the set of all even integers, are isomorphic with the isomorphism $x \mapsto 2x$.

(b) The groups $(\mathbb{R}, +_{\mathbb{R}})$ and $(\mathbb{R}_{>0}, +_{\mathbb{R}})$, are isomorphic with respect to the real exponential map: $x \mapsto exp(x)$.

There are two more definitions that will become important for later:

[12] i.e. a map that is both injective (one to one) and surjective (onto)

Definition 3.8. Let A be a set. An **Endomorphism** is an isomorphism from A to itself. The set of all the Endomorphisms is denoted by $End(A) := \{f : A \to A \,|\, f \text{ is an isomorphism } \}$.

Definition 3.9. We define an **Automorphism** to be an invertable Endomorphism. The set of all the Automorphisms is denoted by $Aut(A) := \{f : A \to A \,|\, f \text{ is an invertable isomorphism } \}$.

It is easy to see that for a set A, $Aut(A) \subset End(A)$.

These sets of maps have some very interesting properties themselves and can be inherited with some extra structure to become vector spaces or even groups.

3.3 Permutations

Another useful and very applicable concept in group theory is permutations. This is one of the concepts that may seem abstract at first, but when understood are fairly simple: It is the way of rearranging the elements of a set.

Definition 3.10. A **permutation** on some set G is a bijective map $f : G \to G$.

Theorem 3.4. *Let the set of all the permutations on some set G be denoted as $Perm(G)$. $Perm(G)$ forms a group under map composition: $(Perm(G), \circ)$*

Proof. We have to prove that the four group axioms are satisfied:

1. Closure: $\forall f, g \in Perm(G) : f \circ g : G \to G \in Perm(G)$.

2. Asscociativity: $\forall f, g, h \in Perm(G) : (f \circ g) \circ h = f \circ (g \circ h)$.

3. The identity element of $Perm(G)$ is the identity map $id_{Perm(G)} : G \to G \in Perm(G)$ defined as $x \mapsto x$ for $x \in G$, since $f \circ id_{Perm(G)} = id_{Perm(G)} \circ f = f$.

4. The inverse of any element $f \in Perm(G)$ is its inverse mapping: $f^{-1} : G \to G \in Perm(G)$, since $f^{-1} \circ f = f \circ f^{-1} = id_{Perm(G)}$.

Lastly, map composition is commutative and thus $(Perm(G), \circ)$ is abelian.■

Note that the number of permutations of n elements is $n!$, thus the order of $(Perm(G), \circ)$ is $|G|!$
Sometimes, the permutation group is also called the **symmetric group** of degree n, noted by S_n.

3.4 Equivalence relations

An important notion in various mathematical fields is the equivalence relation, covering a wide spectrum of objects and helping to classify them in a very useful way.

Definition 3.11. Let S be some set. We define a binary operation on S, denoted by "$\sim$". If the three following properties are satisfied, then "$\sim$" is called an **equivalence relation** on S:

1. Reflexivity: $\forall x \in S : x \sim x$
2. Symmetry: $\forall x, y \in S : x \sim y \iff y \sim x$
3. Transitivity: $\forall x, y, z \in S$: if $x \sim y$ and $y \sim z \Rightarrow x \sim z$

Examples:

(a) A simple example of an equivalence relation is the following: For two people x, y, define $x \sim y :\iff$ (x has the same opinion as y). One can trivially prove that the three properties are satisfied. $\square$

(b) Let $x, y \in \mathbb{R}$. Define $x \sim y :\iff x - y \in \mathbb{Z}$.
It is reflexive as $\forall x \in \mathbb{R} : x - x = 0 \in \mathbb{Z}$.
To prove that it is symmetric, take some $x, y \in \mathbb{R}$ such that $x \sim y \Rightarrow x - y = n \in \mathbb{Z}$. Then, $y - x = -(x - y) = -n \in \mathbb{Z}$, so $y \sim z$.
Now let $x, y, z \in \mathbb{R}$ and suppose that $x \sim y \Rightarrow x - y = n \in \mathbb{Z}$ and $y \sim z \Rightarrow y - z = m \in \mathbb{Z}$. Then $x - z = (x - y) + (y - z) = n + m \in \mathbb{Z}$, so $x \sim z$. Therefore, "$\sim$" is an equivalence relation. $\square$

Now we can define a subset of S, by gathering together all the equivalent elements.

Definition 3.12. Let $\sim$ be an equivalence relation on S. Then for any $x \in S$ we define the set:

$$[x] := \{a \in S \mid x \sim a\}$$

called the **equivalence class** of x.

Theorem 3.5. *For two equivalence classes on a set S, $[x] = [y] \iff x \sim y$.*

Proof. From the definition of an equivalence class, we have $[x] = \{a \in S \,|\, x \sim a\} = [y] = \{a \in S \,|\, y \sim a\}$. From the symmetric property of the equivalence relation, $y \sim a \Rightarrow a \sim y$. Then we have that $x \sim a$ and $a \sim y$, so from the transitivity property, $x \sim y$. The oposite is proven similarly. $\blacksquare$

3.5 The integers modulo n

We now come to a very common concept in abstract algebra, and that is the integers modulo n.

Definition 3.13. Let n be a positive integer. Two numbers a and b are said to be **congruent modulo n** if their difference is divisible by n. Denoted by $a \equiv b \ (mod\, n)$.

The reader should try proving that congruence $mod\, n$ is an equivalence relation on $\mathbb{Z}$.
Since it is an equivalence relation, we can define the **congruence class of a modulo n** as the equivalence class: $[a]_n := \{b \in \mathbb{Z} \,|\, b \equiv a\, (mod\, n)\} = \{b \in \mathbb{Z} \,|\, b - a = kn \text{ for some } k \in \mathbb{Z}\}$. Note that from Theorem 3, $[a]_n = [a']_n \iff a \sim a'$.

Definition 3.14. We define the set of all congruence classes modulo n, as the **integers modulo n**, denoted by[13]:

$$\mathbb{Z}_n := \{[a]_n \,|\, a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \cdots, [n-1]_n\}$$

Now, we define the addition and multiplication maps on $\mathbb{Z}_n$ as:

$$+_n : \mathbb{Z}_n \times \mathbb{Z}_n \to \mathbb{Z}_n$$

$$[a]_n +_n [b]_n := [a +_{\mathbb{Z}} b]_n$$

$$\cdot_n : \mathbb{Z}_n \times \mathbb{Z}_n \to \mathbb{Z}_n$$

$$[a]_n \cdot_n [b]_n := [a \cdot_{\mathbb{Z}} b]_n$$

This gives rise to a rather piculear arithmetic. For example, take $\mathbb{Z}_5 = \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. In this set, one can see that $[1]_5 = [6]_5 = [-4]_5$ and $[3]_5 = [8]_5 = [-2]_5$, since every difference of those numbers is an integer multiple of 5. So, working $(mod\, 5)$: $[1]_5 +_5 [6]_5 = [1+6]_5 = [7]_5 = [2]_5$.

Theorem 3.6. *The integers modulo n form a group under addition.*

Proof.
1. Closure: $\forall\, [a]_n, [b]_n \in \mathbb{Z}_n : [a]_n +_n [b]_n = [a+b]_n \in \mathbb{Z}_n$.

2. Associativity: $\forall\, [a]_n, [b]_n, [c]_n \in \mathbb{Z}_n$:

$$([a]_n +_n [b]_n) +_n [c]_n = [a +_{\mathbb{Z}} b]_n +_n [c]_n = [a+b+c]_n$$

$$\Rightarrow [a]_n +_n [b+c]_n = [a]_n +_n ([b]_n +_n [c]_n)$$

3. The identity element of $\mathbb{Z}_n$ is $[0]_n$, since $\forall\, [a]_n \in \mathbb{Z}_n : [a]_n +_n [0]_n = [a+0]_n = [a]_n$.

[13] generally in abstract algebra, the integers modulo n are denoted by $\mathbb{Z}/n\mathbb{Z}$ because the symbol $\mathbb{Z}_n$ is already given to another algebraic structure. But for our interest we will denote them by $\mathbb{Z}_n$

4. The inverse of any element $[a]_n \in \mathbb{Z}_n$ is $[-a]_n$, since:

$$[a]_n +_n [-a]_n = [a-a]_n = [0]_m$$

We also have the commutative axion satisfied, $\forall [a]_n, [b]_n \in \mathbb{Z}_n : [a]_n +_n [b]_n = [a+b]_n = [b+a]_n = [b]_n +_n [a]_n$.

Therefore, $(\mathbb{Z}_n, +_n)$ is an Abelian group.

Another way to show this, is to notice that $[1]_n$ is a generating element of $(\mathbb{Z}_n, +_n)$:

$$< [1]_n >= \{\underbrace{[1]_n +_n [1]_n +_n \cdots +_n [1]_n}_{\text{n times}} \mid n \in \mathbb{N}\} = \mathbb{Z}_n$$

which makes $(\mathbb{Z}_n, +_n)$ cyclic and thus abelian. $\blacksquare$

4 The relation between Music theory and Group theory

4.1 Pitches in an octave, represented by $(\mathbb{Z}_{12}, +_{12})$

Taking another look at Table 1 while having a background on the integers modulo n, one can see that the musical intervals can be represented by the first twelve integers, that is $\mathbb{Z}_{12}$. Each integer expresses how many semitones are in the chosen interval. So we now strictly define the musical interval between two pitches as their pitch-class interval:

Definition 4.1. For two pitches $a, b \in \mathbb{Z}_{12}$, their **pitch-class**[14] **interval** is defined as $a - b$ $(mod\, 12)$.

Since we proved that $\mathbb{Z}_n$ forms a cyclic group under addition, we have that $(\mathbb{Z}_{12}, +_{12})$ is also a cyclic group, as a subgroup of $(\mathbb{Z}_n, +_n)$. We can also drop the brackets on the elements of $\mathbb{Z}_{12}$ but it is important to remember that these are equivalence classes and not the numbers themselves:

$$\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

And as for the musical interpretation, these are not pitches, but their pitch-classes, since for example "0" can represent C, $B^{\sharp}$, or $D^{\flat\flat}$. From now on, we will use the following definition for denoting chords:

Definition 4.2. A **pitch-class set** is an unordered set of pitch-classes, represented as $[a, b, c]$, where $a, b, c \in \mathbb{Z}_{12}$.

For example an Am chord is $[9, 0, 4]$. This notation helps express chords more generally, regardless of the tonic. In this manner, a X-major (XM) chord is $[x, x + 4, x + 7]$ and a X-minor (Xm) chord is $[x, x + 3, x + 7]$ for some pitch-class $x \in \mathbb{Z}_{12}$. Here, x is often called the **root**.

Let us continue by writing the generating elements of the group, along with the sequence they produce once integer addition $(mod\, 12)$ is acted on them:

$$< 1 >= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 0\} = \mathbb{Z}_{12}$$

which is equivalent to $\{C^{\sharp}, D, D^{\sharp}, E, F, F^{\sharp}, G, G^{\sharp}, A, A^{\sharp}, B, C\}$

$$< 5 >= \{5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 0\} = \mathbb{Z}_{12}$$

which is equivalent to $\{F, A^{\sharp}, D^{\sharp}, G^{\sharp}, C^{\sharp}, F^{\sharp}, B, E, A, D, G, C\}$[15]

$$< 7 >= \{7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, 0\} = \mathbb{Z}_{12}$$

which is equivalent to $\{G, D, A, E, B, F^{\sharp}, C^{\sharp}, G^{\sharp}, D^{\sharp}, A^{\sharp}, F, C\}$[16]

[14] see Appendix D

[15] or $\{F, B^{\flat}, E^{\flat}, A^{\flat}, D^{\flat}, G^{\flat}, B, E, A, D, G, C\}$

[16] or $\{G, D, A, E, B, G^{\flat}, D^{\flat}, A^{\flat}, E^{\flat}, B^{\flat}, F, C\}$

$$< 11 >= \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0\} = \mathbb{Z}_{12}$$

which is equivalent to $\{B, B^\flat, A, A^\flat, G, G^\flat, F, E, E^\flat, D, D^\flat, C\}$

This may seem a bit random at first, but notice that $< 1 >$ is actually the ascending chromatic scale, i.e. playing an octave semitone by semitone:

Figure 7: Ascending chromatic scale, starting from C.

While $< 11 >$ is the descending chromatic scale:

Figure 8: Descending chromatic scale, starting from C.

As for the other two elements, they represent what we call "The circle of fifths" and its inverse, "The circle of fourths".

Figure 9: The circle of fifths, starting from C.

Figure 10: The circle of fourths, starting from C.

These sequences are of most importance in music theory. One can substitute those pitches with their chords (major or minor) to get a very widely used result, in both classical music and jazz[17].

[17] A very famous classical example is Vivaldi's "Winter" from "The Four Seasons" and a jazz one is "Fly me to the moon" by Frank Sinatra.

4.2 Transpositions and Inversions

In subsection 3.3 we talked about group permutations. Defining two very important permutations for $\mathbb{Z}_{12}$ will show how useful they really are in music theory:

Definition 4.3. We define a **Transposition** as the permutation which moves a pitch-class up[18] by n semitones:

$$T_n : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$$
$$x \mapsto x + n \ \ (mod\, 12)$$

A very interesting example of transpositions are the accidentals. One can view them as the following permutations:

$$\sharp \equiv T_1 : \mathbb{Z}_{12} \to \mathbb{Z}_{12} \in Aut(\mathbb{Z}_{12})$$
$$x \mapsto x + 1 \ \ (mod 12)$$

$$\flat \equiv T_{11} : \mathbb{Z}_{12} \to \mathbb{Z}_{12} \in Aut(\mathbb{Z}_{12})$$
$$x \mapsto x + 11 \ \ (mod 12)$$

$$\natural : \mathbb{Z}_{12} \to \mathbb{Z}_{12} \in Aut(\mathbb{Z}_{12})$$

$$x \mapsto \begin{cases} x & \text{if } x \text{ hasn't been shifted a semiton, i.e. is a "white" note } (id_{\mathbb{Z}_{12}} \text{ or } T_0) \\ x+1 \ \ (mod 12) & \text{if } x \text{ has been shifted a semiton down, i.e. has been "flated" } (T_1) \\ x+11 \ \ (mod 12) & \text{ if } x \text{ has been shifted a semiton up, i.e. has been "sharped" } (T_{11}) \end{cases}$$

Definition 4.4. We define an **Inversion** as the permutation which maps a pitch-class to its antisymmetric pitch-class with respect to "0":

$$I : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$$

$$x \mapsto -x \ \ (mod\, 12)$$

Now one sees that the musical terminology: "Transpositions" and "Inversions" is the same as the mathematical terminology: "Translations" and "Reflexions".

We can also construct the map $T_n \circ I \equiv T_n I : \mathbb{Z}_{12} \to \mathbb{Z}_{12}$ which is the generalized inversion, taking a pitch-class, mirroring it with respect to "0" and then moving it up by n semitones:

$$x \mapsto -x + n \ \ (mod\, 12)$$

[18] We only cover the case for a pitch to be moved "up" by a semitone, because moving it "down" by n semitones is the same as moving it up by $12 - n \ \ (mod\, 12)$.

Although this seems to be the case for describing music theory, one faces some problems while trying to map a major chord to its relative minor chord. For example for mapping a CM to a Cm chord you need T_7I:

$$T_7I([0,4,7]) = [-0+7,-4+7,-7+7] = [7,3,0] = [0,3,7]$$

But for mapping DM to Dm you need $T_{11}I$:

$$T_{11}I([2,6,9]) = [-2+11,-6+11,-9+11] = [9,5,2] = [2,5,9]$$

It is imposible to construct a unique permutation that maps XM to Xm. That is the reason we present the Uniform Triadic Transformations.

4.3 The Uniform Triadic Transformations

In 2002, J.Hook released [2] where he introduced the UTTs. Using them, one can analyze whole musical sheets while maintaining the mathematical formalization. It is a compact and useful way of expressing all the previous concepts and improving them.

Definition 4.5. We define a **triad** to be an ordered pair $\Delta = (r, \sigma)$, where $r \in \mathbb{Z}_{12}$ is the **root** of the triad and $\sigma \in \{+,-\}$ is the **sign** which denotes the mode of the triad, i.e. major or minor.

Examples:

(a) $\Delta = (2,-)$ represents the Dm triad $[2,5,9]$, while $\Delta = (2,+)$ the DM triad $[2,6,9]$.

(b) $\Delta = (7,-)$ represents the Gm triad $[7,10,2]$, while $\Delta = (7,+)$ the GM triad $[7,11,2]$.

Theorem 4.1. *Let $\Gamma := \{\Delta = (r,\sigma) \,|\, r \in \mathbb{Z}_{12}$ and $\sigma \in \{+,-\}\}$ be the set of all the triads. The set Γ together with the multiplication[19] defined by:*

$$\Delta_1 \cdot \Delta_2 = (r_1,\sigma_1)\cdot(r_2,\sigma_2) := (r_1 +_{12} r_2, \sigma_1\sigma_2)$$

forms an abelian group $(\Gamma, \cdot)$.

Proof. It is clear that $(\Gamma,\cdot) \cong (\mathbb{Z}_{12}\times\mathbb{Z}_2, \bullet)$, where in the group $\mathbb{Z}_{12}\times\mathbb{Z}_2$ the operation is defined as:

$$\bullet : \mathbb{Z}_{12}\times\mathbb{Z}_2 \to \mathbb{Z}_{12}\times\mathbb{Z}_2$$
$$(x_1,y_1)\bullet(x_2,y_2) := (x_1 +_{12} x_2, y_1 +_2 y_2)$$

so it is a cyclic group itself. ■

[19] Where the sign multiplication follows the rules: $+\cdot+=+$, $+\cdot-=-\cdot+=-$, $-\cdot-=+$

Definition 4.6. We define a **Triadic Transformation** to be a permutation of $(\Gamma, \cdot)$, i.e. a bijection from Γ to itself.

Note that the TTs as permutations of a group (or a set), can be formed into a new group using map composition as shown in Theorem 3.4.

Definition 4.7. A **Uniform Triadic Transformation** (UTT) is a TT that consists of three parts, noted $U =< \sigma_U, t^+, t^- >$ where $t^+, t^- \in \mathbb{Z}_{12}$, $\sigma \in \{+, -\}$ and acting on a triad $\Delta = (r, \sigma)$ as:

$$(r, \sigma) \mapsto \begin{cases} (r + t^+, \sigma_U) & \text{if } \Delta \text{ is a major triad} \\ (r + t^-, \sigma_U) & \text{if } \Delta \text{ is a minor triad} \end{cases}$$

A UTT is said to be **mode-preserving** if it does not change the mode of the triad that is acted on. If it does change it, we say that it is **mode-reversing**.

Let some examples clear this up:

Examples:

(a) Let $U =< -, 4, 3 >$ and $\Delta = (2, +)$. $U\Delta = (6, -)$.

(b) Let $U =< +, 2, 5 >$ and $\Delta = (9, -)$. $U\Delta = (2, +)$.

(c) Let $U =< -, 3, 4 >$ and $\Delta = (0, -)$. $U\Delta = (4, -)$.

Theorem 4.2. *Let* $\mathscr{U} := \{U : \Gamma \to \Gamma \,|\, U \text{ is a } UTT\}$ *be the set of all the* UTT*s. Equiping it with map composition, it forms a non-abelian group* $(\mathscr{U}, \circ)$.

Proof.
1. Closure: $\forall\ U_1 =< \sigma_1, t_1^+, t_1^- >, U_2 =< \sigma_2, t_2^+, t_2^- > \in \mathscr{U} : U_1 \circ U_2 = U_2 U_1 =< \sigma_1\sigma_2\,,\, t_1^+ +_{12} t_2^+\,,\, t_1^- +_{12} t_2^- > \in \mathscr{U}$.

2. Associativity: Both addition $mod12$ and sign multiplication are associative, thus associativity holds for UTTs.

3. The identity element of and UTT is $< +, 0, 0 >$, since $\forall\ U =< \sigma\,, t^+\,, t^- > \in \mathscr{U}$: $< \sigma\,, t^+\,, t^- >< +, 0, 0 >=< \sigma\,, t^+\,, t^- >$

4. The inverse element of $\mathscr{U}$ depends on wether the UTT is mode-preserving or mode-reserving. For a mode-preserving UTT $< +, t^+, t^- >$, the inverse is $< +, -t^+, -t^- >$ and for a mode-reversing UTT $< -, t^+, t^- >$, the inverse is $< -, -t^+, -t^- >$

The group $(\mathscr{U}, \circ)$ is non-abelian. ■

UTTs are used constantly, producing harmonic chains, i.e. patterns of chords, or even key sequences. An example of that is the Well-Tempered Clavier by Bach, that uses the $UTT < -, 0, 1 >$ to produce the key of the next prelude, starting from C. They are also widely used in his Fugues, where a musical theme is introduced by a voice and then, having acted some transposition on it, is repeated by another one.

4.4 The neo-Riemannian group

In the early 19th-century, H.Riemann[20] developed three particular transformations, the neo-Riemannian operations which can express a wide variety of the classical harmony. After Riemann, D.Lewin extended this idea in [10], in the context of transformation theory. In this subsection, we will review the neo-Riemannian operations as UTTs and define the so-called neo-Riemannian group.

The main idea of the neo-Riemannian transformations is changing one note of a triad to produce another triad. The three transformations are the following:

P: which maps a triad into its parallel, that is reversing its mode by moving its third a semitone down if major, or moving its third a semitone up if minor.

R: which maps a triad into its relative[21] by moving its fifth a tone up if major, or moving its root a tone down if minor.

L: which maps a triad into its leading tone exchange, by moving the root a semitone down if major, or by moving the fifth a semitone up if minor .

We can express them as UTTs in the following manner:

$$P =< -, 0, 0 >$$

$$R =< -, 9, 3 >$$

$$L =< -, 4, 8 >$$

The reader can check that for a constant triad, P preserves the perfect fifth, R preserves the major third and L preserves the minor third.

Definition 4.8. We define the **neo-Riemannian group** as the subgroup of $\mathscr{U}$ which consist of the set of UTTs generated by P, R and L :

$$\mathscr{R} :=< P, R, L > \subset \mathscr{U}$$

and its elements are in the form $< \sigma_U, t^+, t^- >$ where $t^+ +_{12} t^- = 0$

[20] not to be confused with the famous mathematician Bernhard Riemann (1826 – 1866)

[21] We refer to two scales (or in this instance chords) as relatives if they have the same key signatures. For example: CM and Am, FM and Dm, $D^\flat M$ and $B^\flat m$ and so on

4.5 Modern Studies

It can be proven that the mode-preserving neo-Riemannian tranformations form an abelian subgroup of $\mathscr{R}$. This is where the geometric aspect of the musical chords arrises (in [6]), as the transformations $R \circ L$, $P \circ L$ and $P \circ R$ can be related to a 3-torus $\mathbb{T}^3 := \mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^1$. For those who are wondering how is this even possible, Tymoczko provides a notion of continuity by considering chords as points in geometrical spaces and voice leadings as lines between them. This idea is based at the fact that the human ear has logarithmic perception of pitches, more preciesly for a given frequency f, we assigne a real number as:

$$p = 69 + 12\, log_2 \left(\frac{f}{440} \right) \tag{4.5.1}$$

which provides a linear space, called the **pitch space** and it preserves all the previous remarks (octaves have a size of 12 and semitones a size of 1). Adding the pitch classes to the image [22], the resulting space is a quotient space, in particular the so called "orbit space"[23] $\mathbb{R}/\mathbb{Z}_{12}$ which is isomorphic[24] to $\mathbb{S}^1$. So if a single pitch class $p \in \mathbb{R}/\mathbb{Z}_{12}$, then pitch-class sets, such as chords, must lie in $(\mathbb{R}/\mathbb{Z}_{12})^n \cong \underbrace{\mathbb{S}^1 \times \ldots \mathbb{S}^1}_{\text{n times}} = \mathbb{T}^n$.

[22] as described in Appendix D

[23] From here on more advanced notions of group theory are used. We consider two points x, y equivalent if they lie in the same orbit, i.e. $x \sim y :\Leftrightarrow \exists\, a \in \mathbb{Z}_{12} \,:\, a +_{12} x = y$, where what we call orbits, are the equivalent classes of "$\sim$". The set $\mathbb{R}/\mathbb{Z}_{12}$ is the set of all the orbits, or all such equivalence classes for the real numbers.

[24] with the isomorphism $\mathbb{R}/\mathbb{Z}_{12} \ni x \mapsto exp(2\pi i/x) \in \mathbb{S}^1$

A Musical Notation

The modern musical notation is describing both the duration and the pitch of a note. The table shows the symbols of duration of a pitch and the figure below how they are playced on the pentagramm for denoting the note:

Table 2

Symbol	Name	Duration (in 4/4 tempo)
𝅝	Whole	4/4
𝅗𝅥	Half	2/4
♩	Quarter	1/4
♪	Eighth	1/8
⋮	⋮	⋮

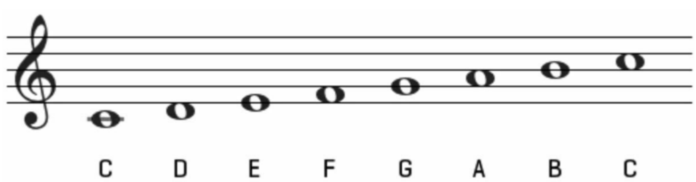
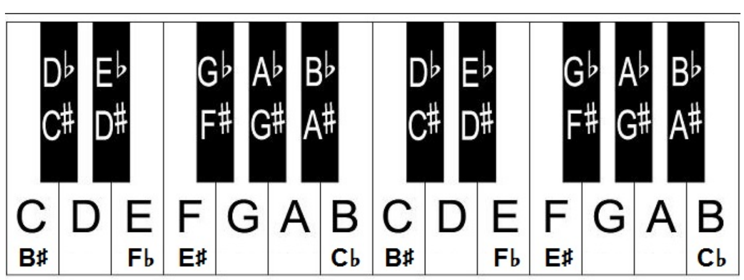


Figure 11: An octave begining at the central C

B Pitches on the Piano



C Proof of Example (b) of the Theorem 3.2

We can construct the map

$$f : \mathbb{C}_{\neq 0} \to \mathbb{S}^1 \times \mathbb{R}_{>0}$$
$$z \mapsto \left(\frac{z}{|z|}, |z| \right)$$

and show that it is an isomorphism.

For any $z \in \mathbb{C}_{\neq 0}$, using the polar representation of a complex number, we write $z = re^{i\theta}$ having $r \in \mathbb{R}_{>0}$ and $e^{i\theta} \in \mathbb{S}^1$

Now let $z_1, z_2 \in \mathbb{C}_{\neq 0}$ such that:

$$f(z_1) = f(z_2) \Rightarrow \left(\frac{z_1}{|z_1|}, |z_1| \right) = \left(\frac{z_2}{|z_2|}, |z_2| \right)$$
$$\Rightarrow \begin{cases} \frac{z_1}{|z_1|} = \frac{z_2}{|z_2|} \\ |z_1| = |z_2| \end{cases}$$
$$\Rightarrow z_1 = z_2$$

so f is injective.

Now, consider a point $(e^{i\theta}, r) \in \mathbb{S}^1 \times \mathbb{R}_{>0}$. We use the map $f^{-1} : \mathbb{S}^1 \times \mathbb{R}_{>0} \to \mathbb{C}_{\neq 0}$ as:

$$(e^{i\theta}, r) \mapsto re^{i\theta}$$

To prove that this is the inverse of f:

$$f^{-1}(f(re^{i\theta})) = f^{-1}(\frac{re^{i\theta}}{r}, r) = re^{i\theta}$$
$$\Rightarrow f^{-1} \circ f = id_{\mathbb{C}_{\neq 0}}$$

and

$$f(f^{-1}(e^{i\theta}, r)) = f(re^{i\theta}) = (\frac{re^{i\theta}}{r}, r) = (e^{i\theta}, r)$$
$$\Rightarrow f \circ f^{-1} = id_{\mathbb{S}^1 \times \mathbb{R}_{>0}}$$

This map takes every point on the cylinder $\mathbb{S}^1 \times \mathbb{R}_{>0}$ and maps it into a unique non-zero point on the complex plane. This proves that f is also surjective.

We've got a bijective map from $\mathbb{C}_{\neq 0} \to \mathbb{S}^1 \times \mathbb{R}_{>0}$, but for it to be an isomorphism, we,ve got to prove that its a homomorphism first, i.e. it preserves complex multiplication:

$$f(z_1 \cdot z_2) = f(r_1 e^{i\theta_1} r_2 e^{i\theta_2}) = f(r_1 r_2 e^{i\theta_1} e^{i\theta_2})$$
$$= \left(\frac{|r_1 r_2 e^{i\theta_1} e^{i\theta_2}|}{r_1 r_2}, r_1 r_2 \right) = (e^{i\theta_1} e^{i\theta_2}, r_1 r_2)$$

$$= (e^{i\theta_1}, r_1) \cdot (e^{i\theta_2}, r_2) = f(r_1 e^{i\theta_1}) \cdot f(r_2 e^{i\theta_2})$$
$$\Rightarrow f(z_1 \cdot z_2) = f(z_1) \cdot f(z_2)$$

So we proved that f is a bijective homomorphism, which is an isomorphism [25] from $\mathbb{C}_{\neq 0}$ to $\mathbb{S}^1 \times \mathbb{R}_{>0}$. Thus,

$$(\mathbb{C}_{\neq 0}, \cdot_{\mathbb{C}}) \cong (\mathbb{S}^1 \times \mathbb{R}_{>0}, \cdot) \ \blacksquare$$

D Pitch classes as equivalence relations

When we refair to a "pitch class" we are actually describing an equivalence class. Consider the following equivalence relation[26] :

$$a \sim b :\Leftrightarrow a - b = 12n \ \ , n \in \mathbb{Z}$$

This labels two pitches equivelent if their distance is an integer multiple of an octave. Now let $x \in \mathbb{Z}_{12}$. The Pitch class is defined as:

$$[x] := \{a \in \mathbb{Z}_{12} \mid x \sim a\} = \{a \in \mathbb{Z}_{12} \mid x - a = 12n \ \ , n \in \mathbb{Z}\}$$

So, for example, all the $G^\sharp$ s are considered "the same" since they are equivalent.

In other words, this uses the fact that all the keys are actually one repeating octave, making the mathematical description more clear.

[25] We also have that for every element $z \in \mathbb{C}_{\neq 0}$ can be written as $z = \frac{z}{|z|} \cdot |z|$ and $\mathbb{S}^1 \cap \mathbb{R}_{>0} = \{1\} = \{e_{\mathbb{C}_{\neq 0}}\}$ so $\mathbb{C}_{\neq 0} = \mathbb{S}^1 \bigoplus \mathbb{R}_{>0}$.

[26] very simillar to Example (b) in 3.4

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